

Communicating with Data – Text as Data

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Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Plan for Today

- Mid-Semester Project wrap-up
- What is text data?
- Why visualize text?
- Techniques
- Activity (time permitting)

Mid-Semester Project Data Exploration

- Download the mid-semester project dataset from Slack
- Explore the data using (simple) visualizations
- Identify three interesting things in the data

Mid-Semester Project Data Exploration

- Pair up with someone across the room
- Share the three interesting things you found
- What new analysis questions do you have?

Mid-Semester Project Data Exploration

- Class-wide share:
 - What did you find in the data?
 - What new questions do you have?

What is text data?

- Documents
 - Articles, books and novels
 - E-mails, web pages, blogs
- Text snippets
 - Tweets, SMS messages
 - Tags, comments, profiles
- And more...
 - Computer programs, logs
 - Collections of documents
 - This slide!

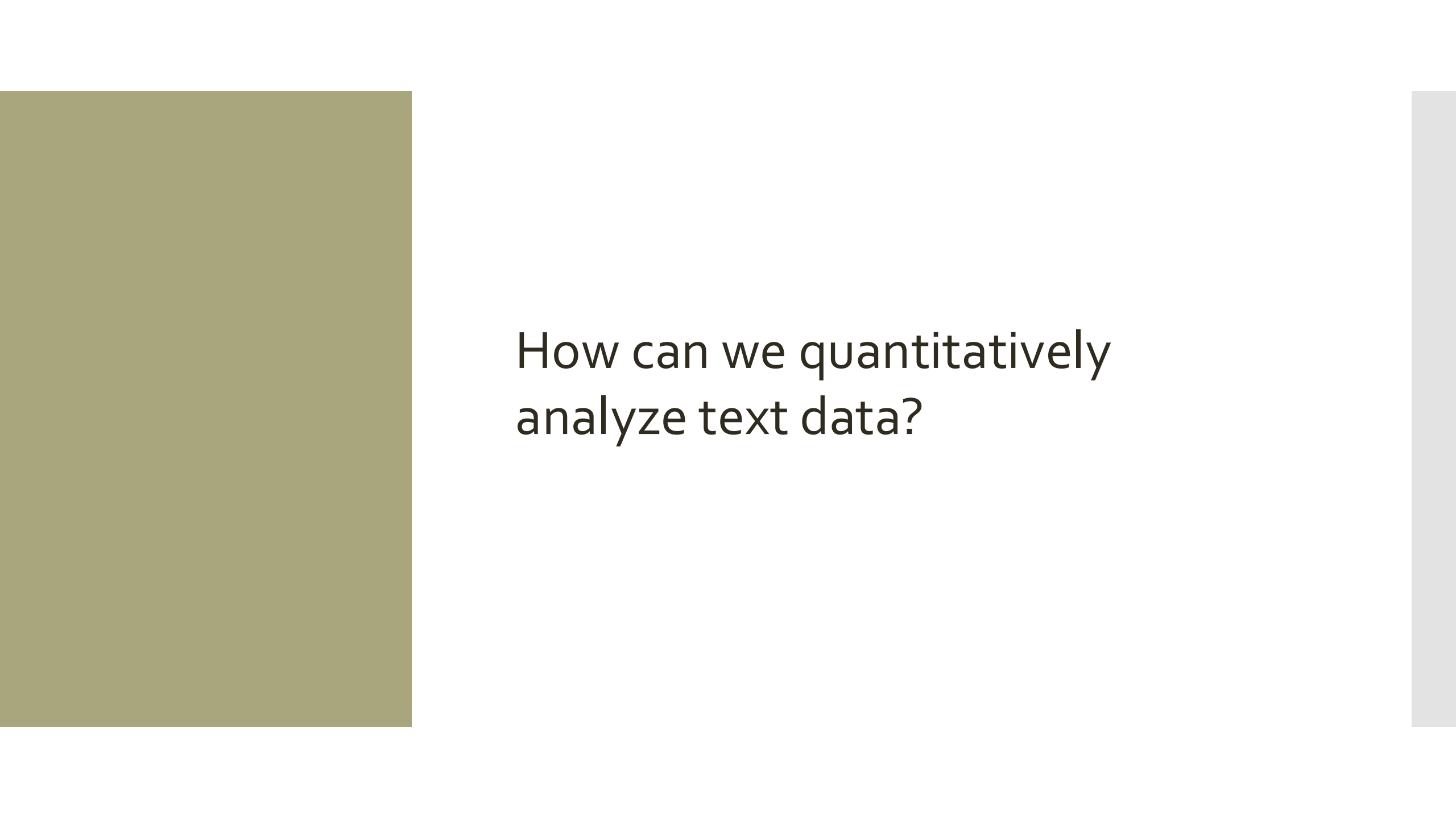


Overview

- **Question:** what are some **characteristics** of text data?
- **Answer:**
 - Often high dimensional (over **228,000** words in OED)
 - Packed with meaning and relationships:
 - **Correlations:** Hong Kong, San Francisco, Bay Area
 - **Order:** April, February, January, June, March, May
 - **Membership:** Tennis, Running, Swimming, Hiking, Piano
 - **Hierarchy,** antonyms & synonyms, entities, ...

Why visualize text data?

- **Understand** – read a document
- **Summarize** – get the “gist” of a document
- **Cluster** – group together similar contents
- **Quantify** – convert to numerical measures
- **Correlate** – compare patterns in text to those in other data, e.g., test scores with conversations on social media

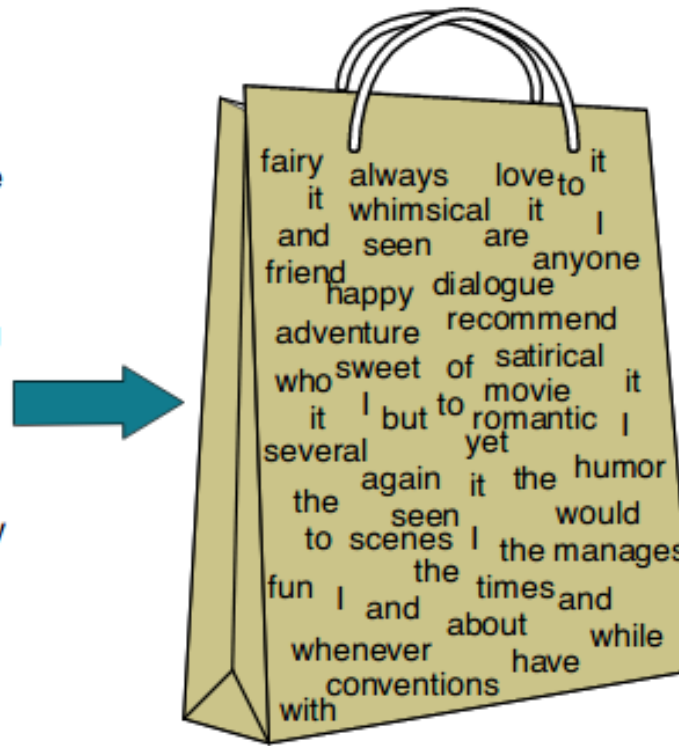


How can we quantitatively
analyze text data?

“Bag of words” model

- Ignore ordering relationships within the text
- Document is represented as a vector of terms and weights
- Grammar, context, and order are ignored

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

“Bag of words” model

- Comparing documents → comparing vectors
- Process:
 - Identify vocabulary, i.e. set of all words in all documents
 - For each document, create a vector of counts of each vocabulary word
 - Compare vectors
- Ex.
 - Sentence 1: “I love cats.”
Sentence 2: “I love dogs.”
 - Vocabulary: [“I”, “love”, “cats”, “dogs”]
 - Sentence 1 vector: [1, 1, 1, 0]
Sentence 2 vector: [1, 1, 0, 1]

“Bag of words” model

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- Ex.
 - Sentence 1: “I love cats.”
 - Sentence 2: “I love dogs.”
 - Sentence 3: “I love cats and dogs.”
- Vocabulary: [“I”, “love”, “cats”, “dogs”, “and”]
- Sentence 1 vector: [1, 1, 1, 0, 0]
- Sentence 2 vector: [1, 1, 0, 1, 0]
- Sentence 3 vector: [1, 1, 1, 1, 1]

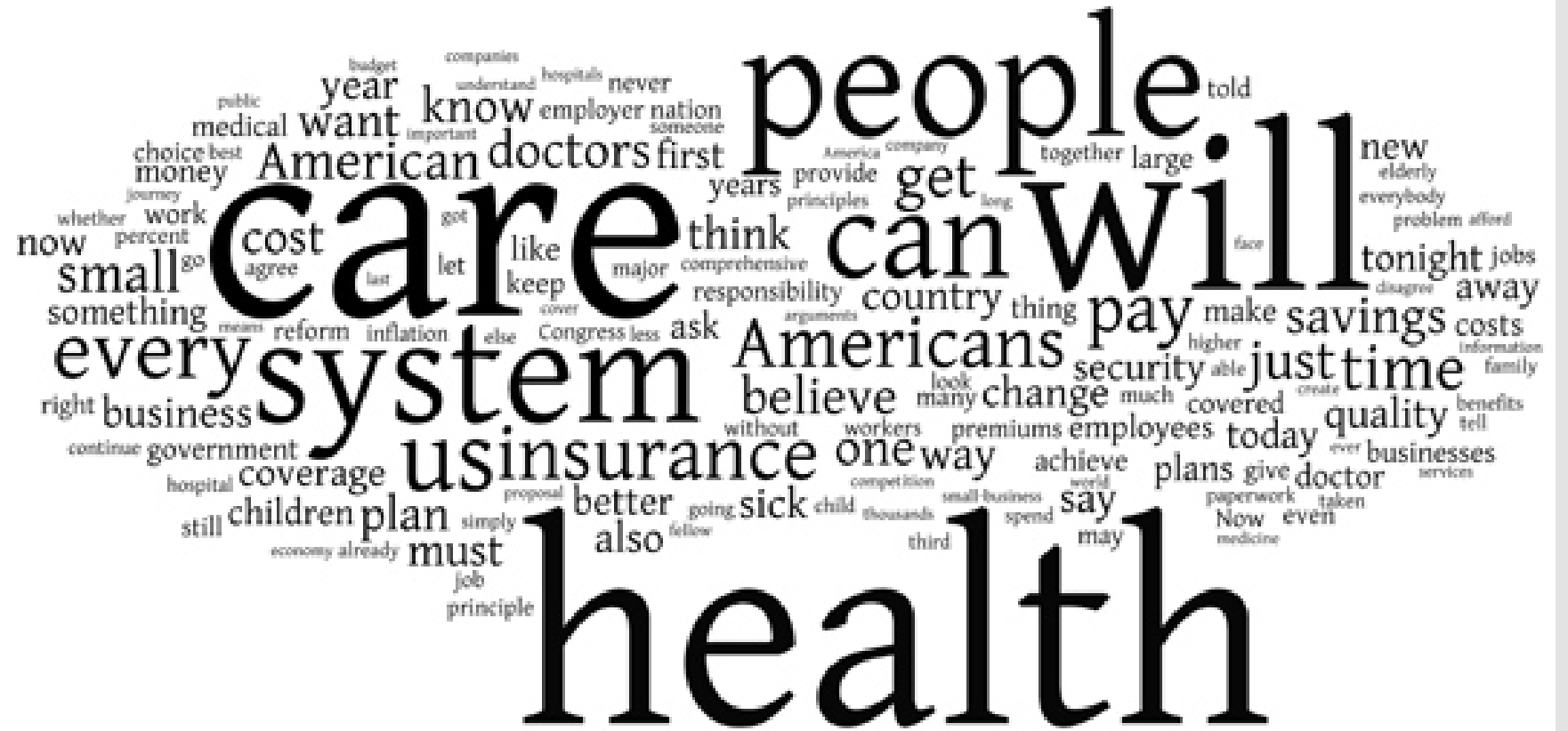
Example: health care reform

- Recent history
 - Initiatives by President Clinton
 - Overhaul by President Obama
- Text data
 - News articles
 - Speech transcriptions
 - Legal documents



New York
Times: Obama
2009

New York Times: Clinton 1993



Obama 2009

Clinton 1993



Rep. Charles Boustany of
Louisiana 2009

Word clouds

- Strengths
 - Familiar to many people
 - Can help with “gisting” and initial query formation
- Weaknesses
 - Does not show the structure of the text
 - Sub-optimal visual encoding (position is not meaningful)
 - Inaccurate size encoding (long words are bigger)
 - May not facilitate comparison (unstable layout)
 - Term frequency **may not be meaningful**

Weighting words

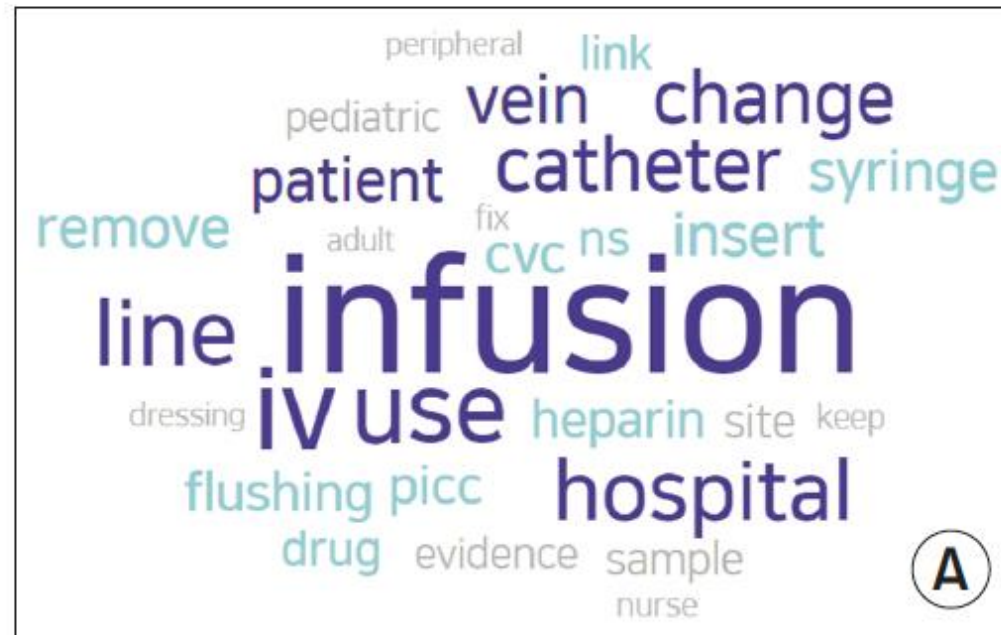
- Term Frequency

$tf_{td} = \#$ of times term t appears in document d

- TF-IDF: Term Frequency by Inverse Document Frequency

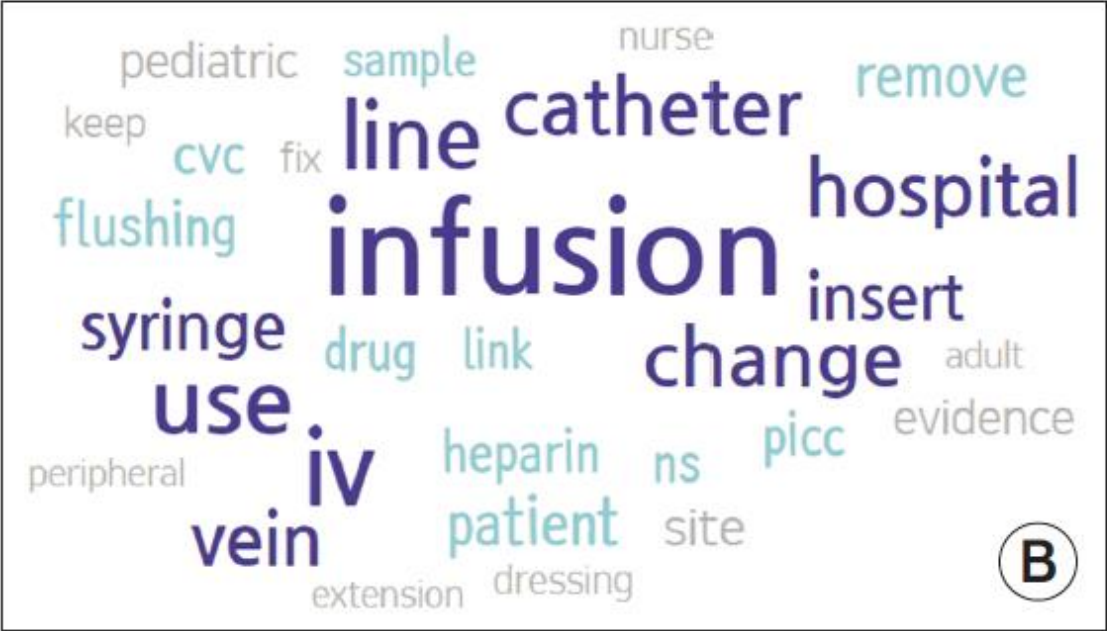
$tf-idf_{td} = \frac{\# \text{ of times term } t \text{ appears in document } d}{\# \text{ of times term } t \text{ appears in all documents}}$

Frequency example



*Keyword Network Analysis of Infusion Nursing
from Posts on the Q&A Board in the Intravenous
Nurses Caf.*

TF-IDF example

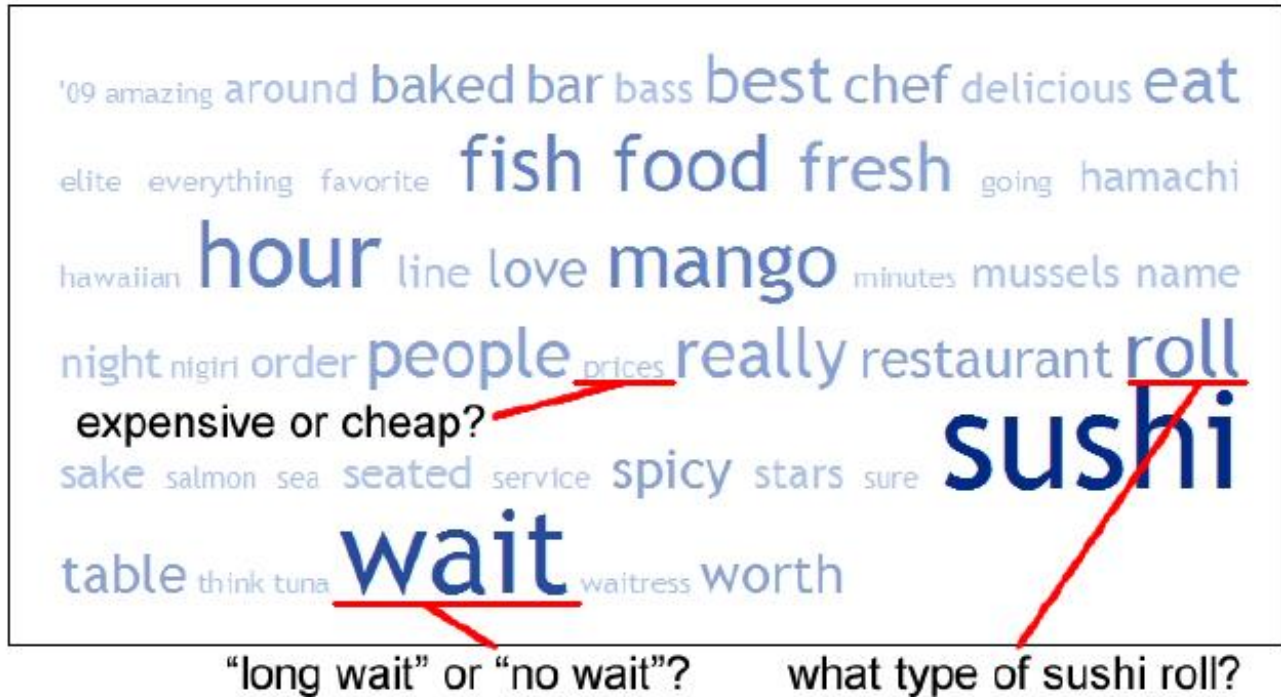


Keyword Network Analysis of Infusion Nursing from Posts on the Q&A Board in the Intravenous Nurses Caf.

Limitations of frequency statistics

- Often favors frequent (TF) or rare (IDF) terms
 - Still not clear that these provide best description
- A “bag of words” ignores additional information
 - Grammar / part-of-speech
 - Position within document
 - Recognizable entities
- Typically focus on unigrams (single terms)

Example: Yelp reviews



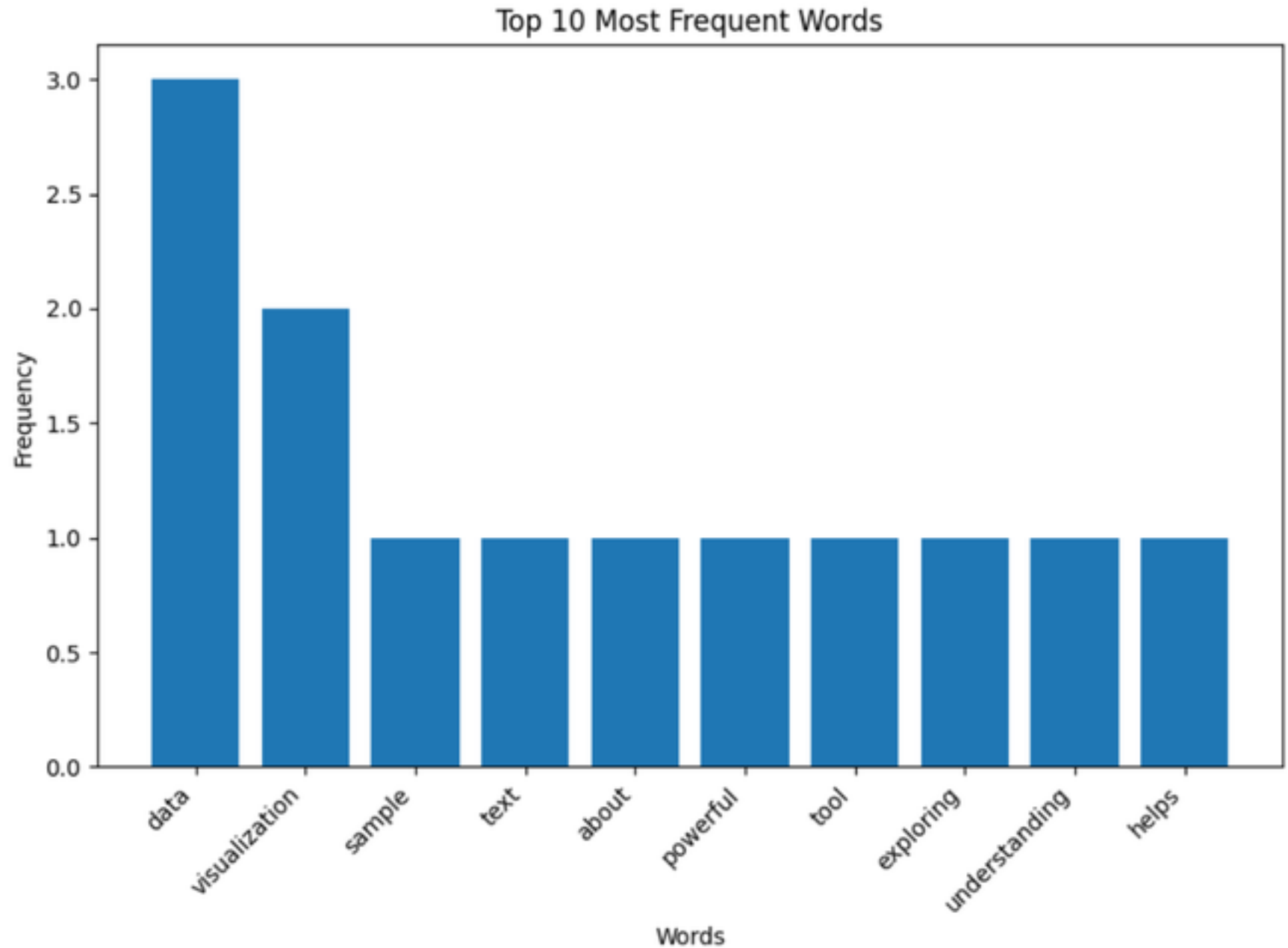
Example: Yelp reviews



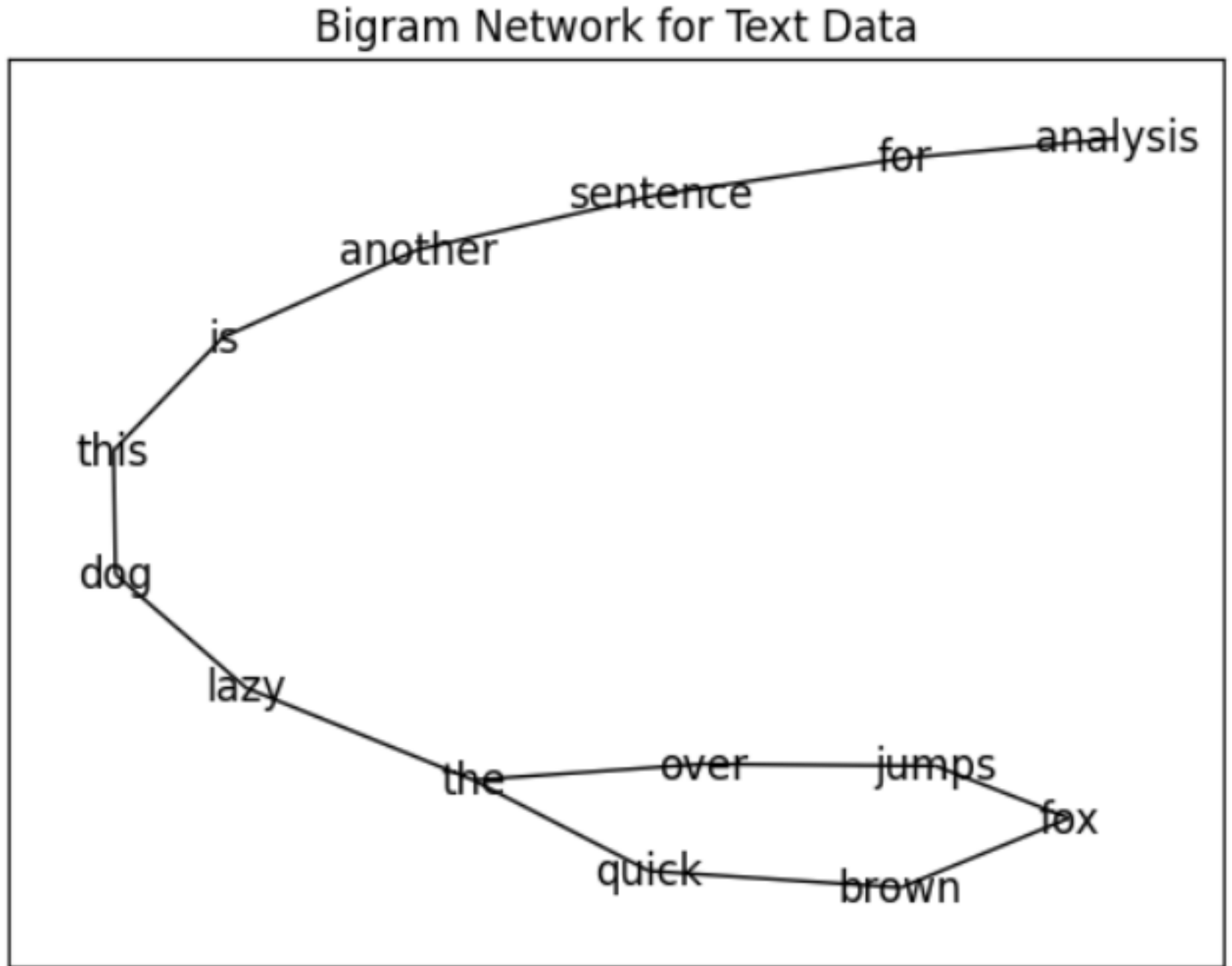
Discussion

What are some other ways we might
visualize text data?

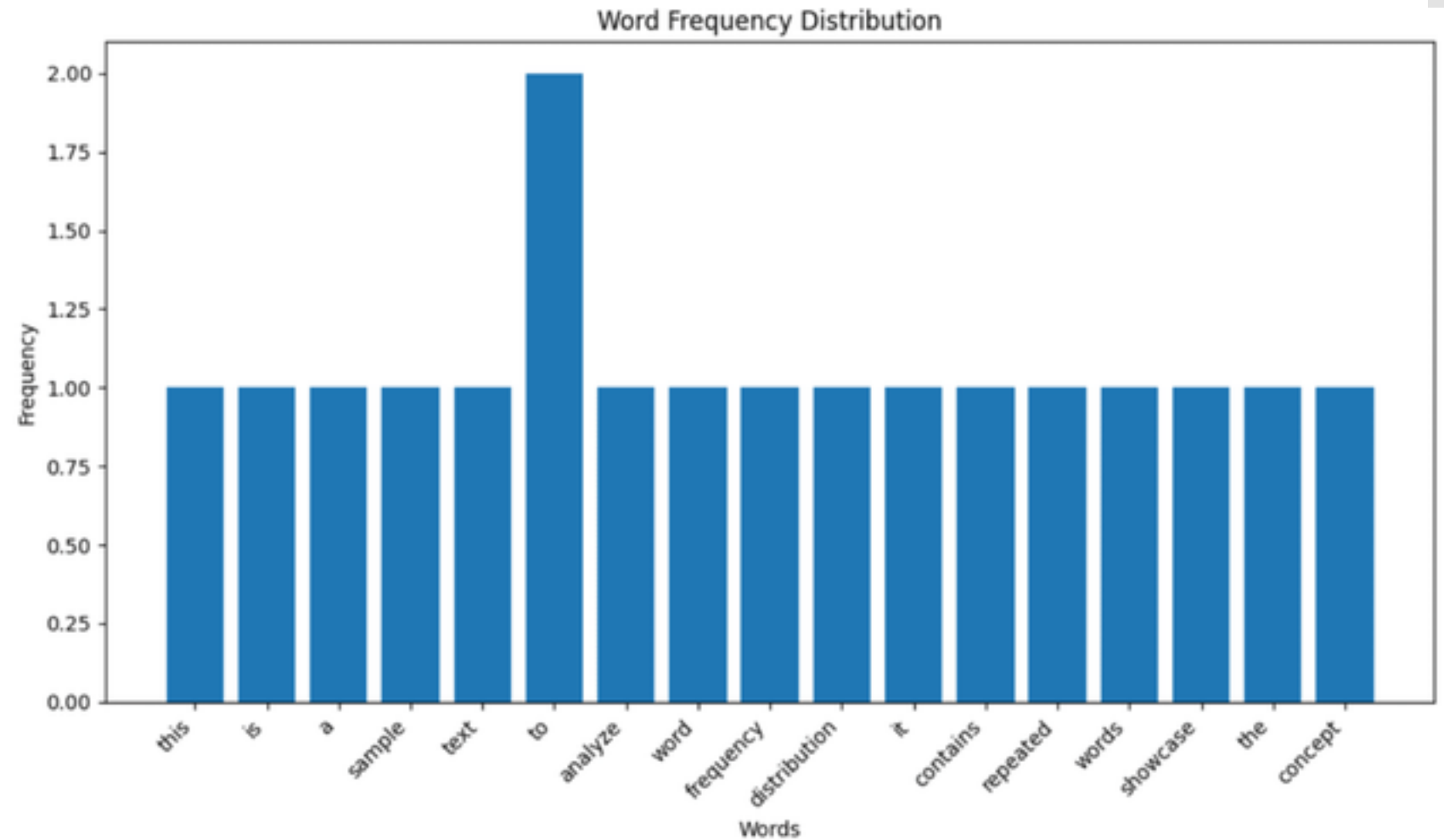
Bar Charts



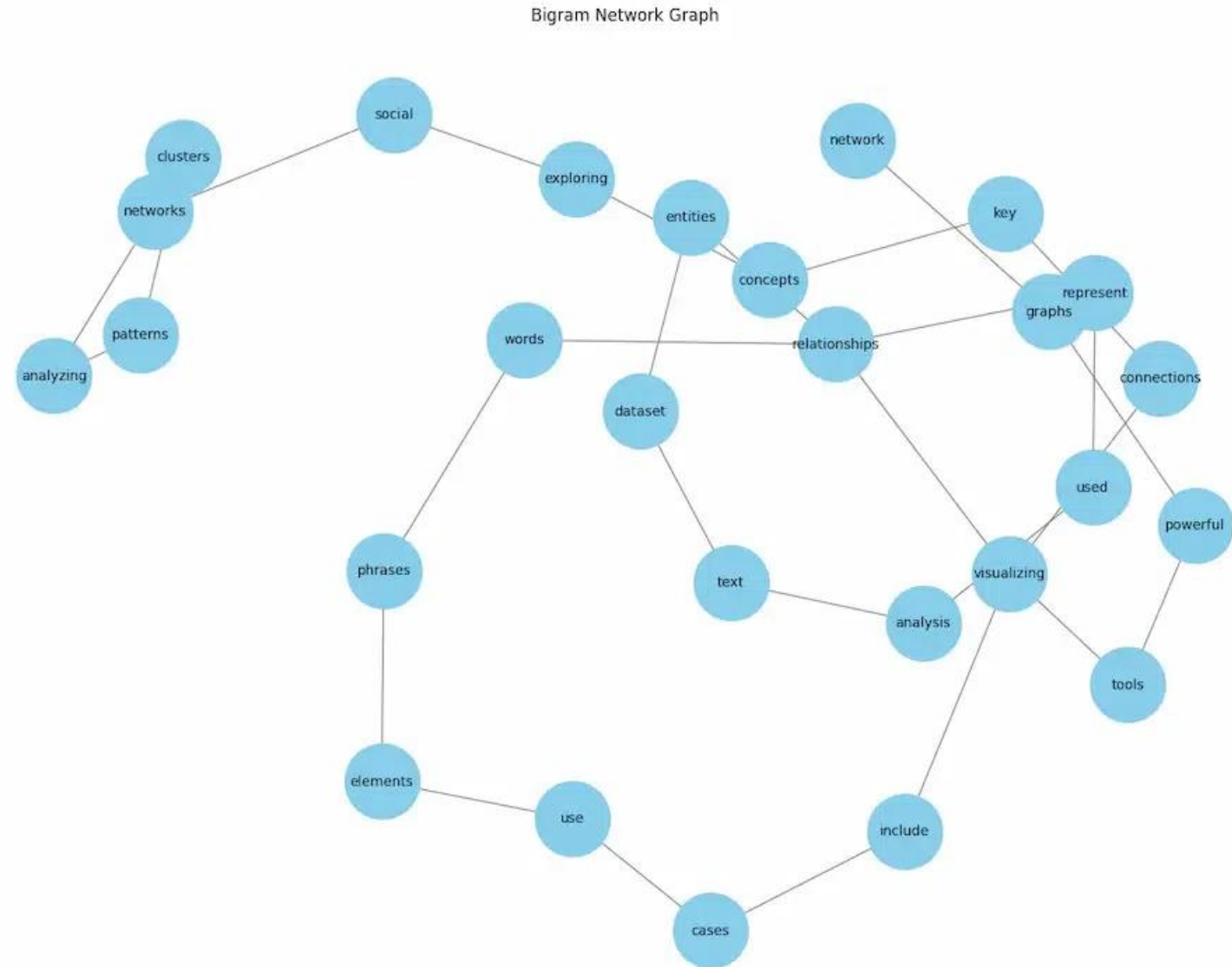
Bigram Network



Word Frequency Distribution Plot



Network Graphs



Activity

- Explore other visualization options. Find one of interest to you, and be ready to share it with the class.
- Some places to look:
 - <https://public.tableau.com/app/discover/viz-of-the-day>
 - <https://observablehq.com/@d3/gallery>
 - <https://r-graph-gallery.com/>
 - <https://informationisbeautiful.net/>
 - <https://python-graph-gallery.com/best-python-chart-examples/>
 - <https://www.census.gov/dataviz/>
 - <https://www.tableau.com/viz-gallery>
 - Anywhere you want to look!